

1

2

3

4

5

6

7

8

A

B

C

D

E

F

power_input

File: power_input.kicad_sch

boost_charger

File: boost_charger.kicad_sch

cap_bank

File: cap_bank.kicad_sch

pulse_switch

File: pulse_switch.kicad_sch

flyback

File: flyback.kicad_sch

sense_gates

File: sense_gates.kicad_sch

mcu_status

File: mcu_status.kicad_sch

OmniMarble

Sheet: /
File: omnimarble-driver.kicad_sch

Title: **OmniMarble 60V SELV coilgun driver**

Size: A3

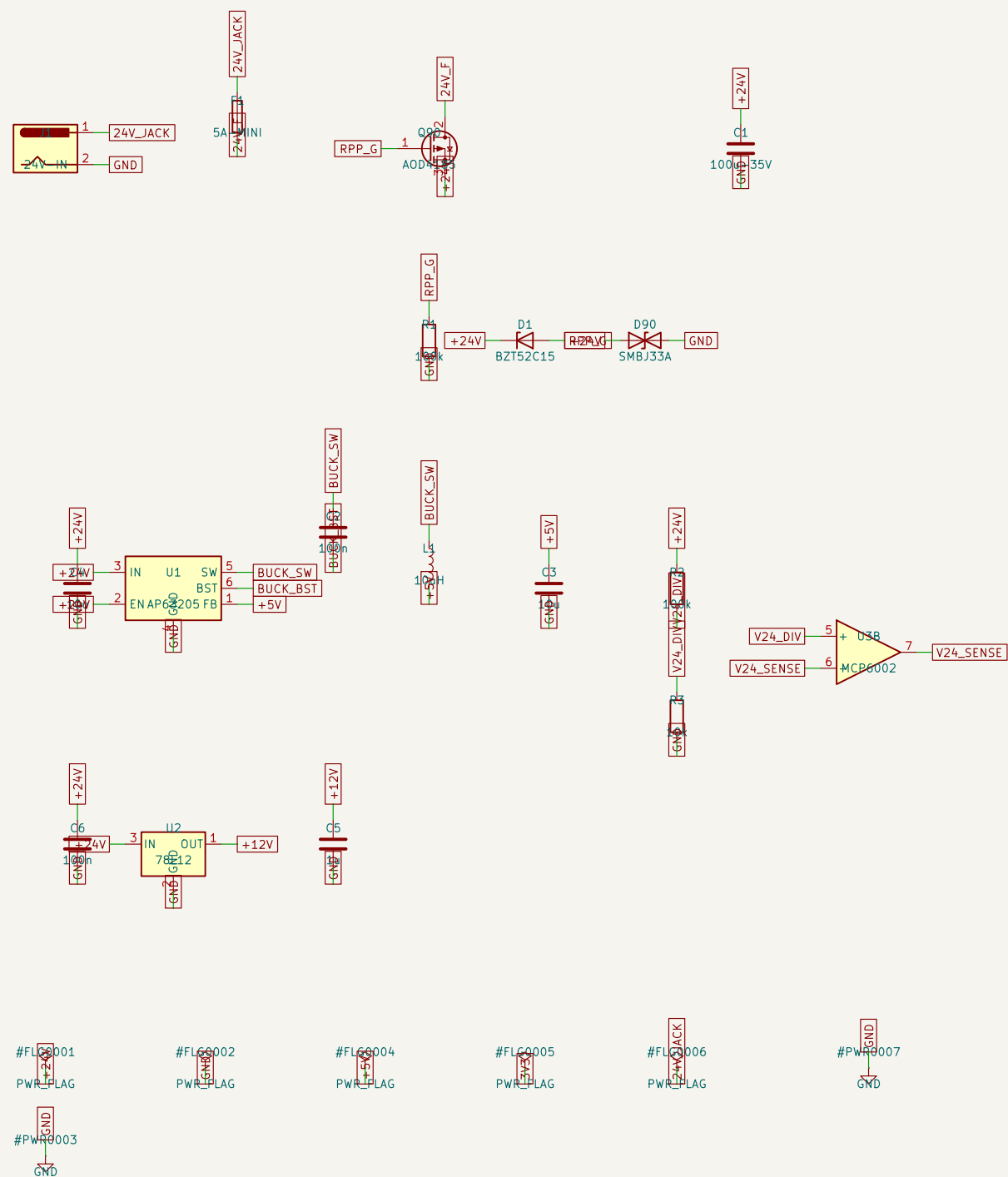
Date:

Rev:

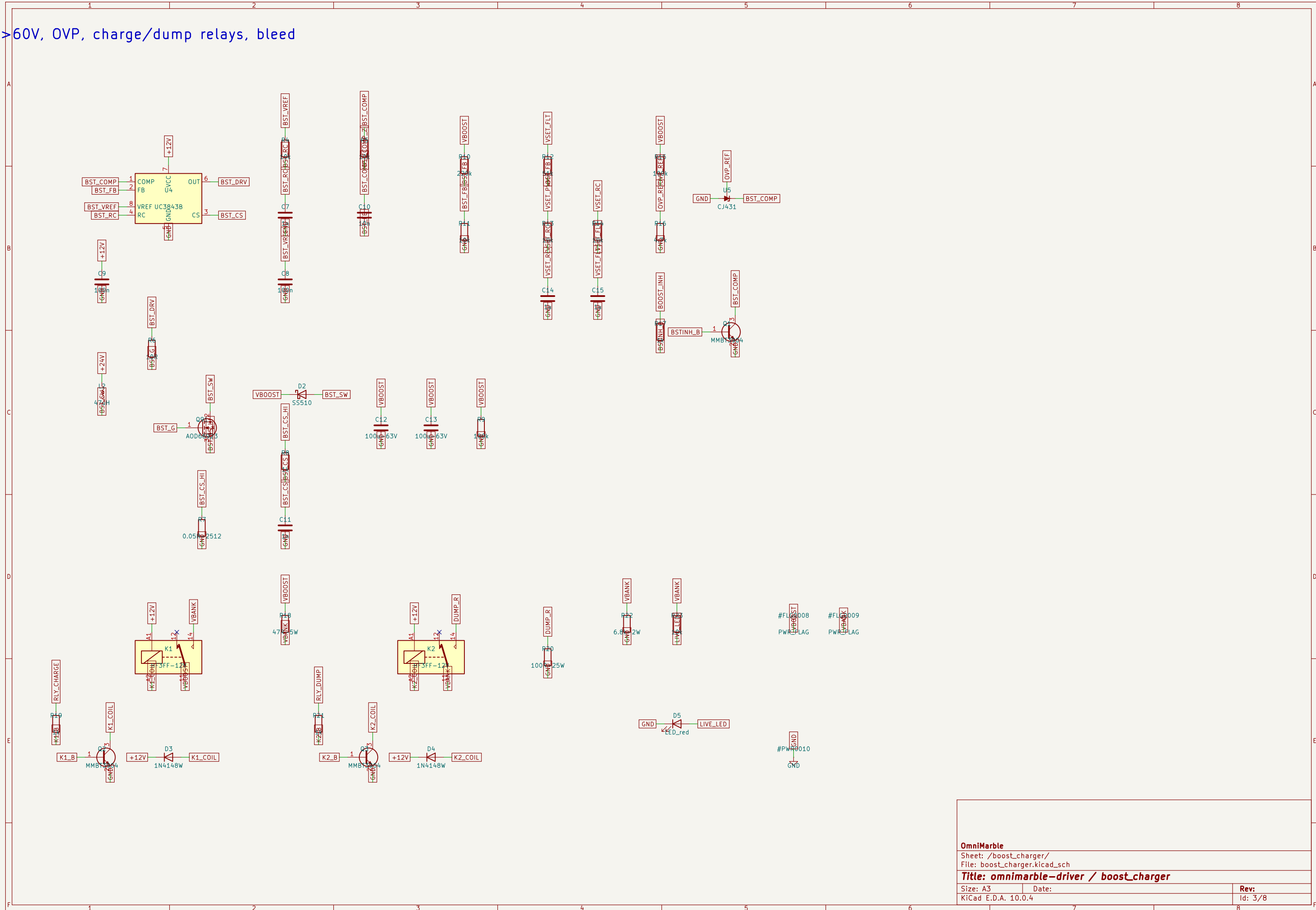
KiCad E.D.A. 10.0.4

Id: 8/8

out, protection, 5V buck, 12V rail



4V→60V, OVP, charge/dump relays, bleed



OmniMarble

Sheet: /boost_charger/
File: boost_charger.kicad_sch

Title: omnimarble-driver / boost_charger

Size: A3

Date:

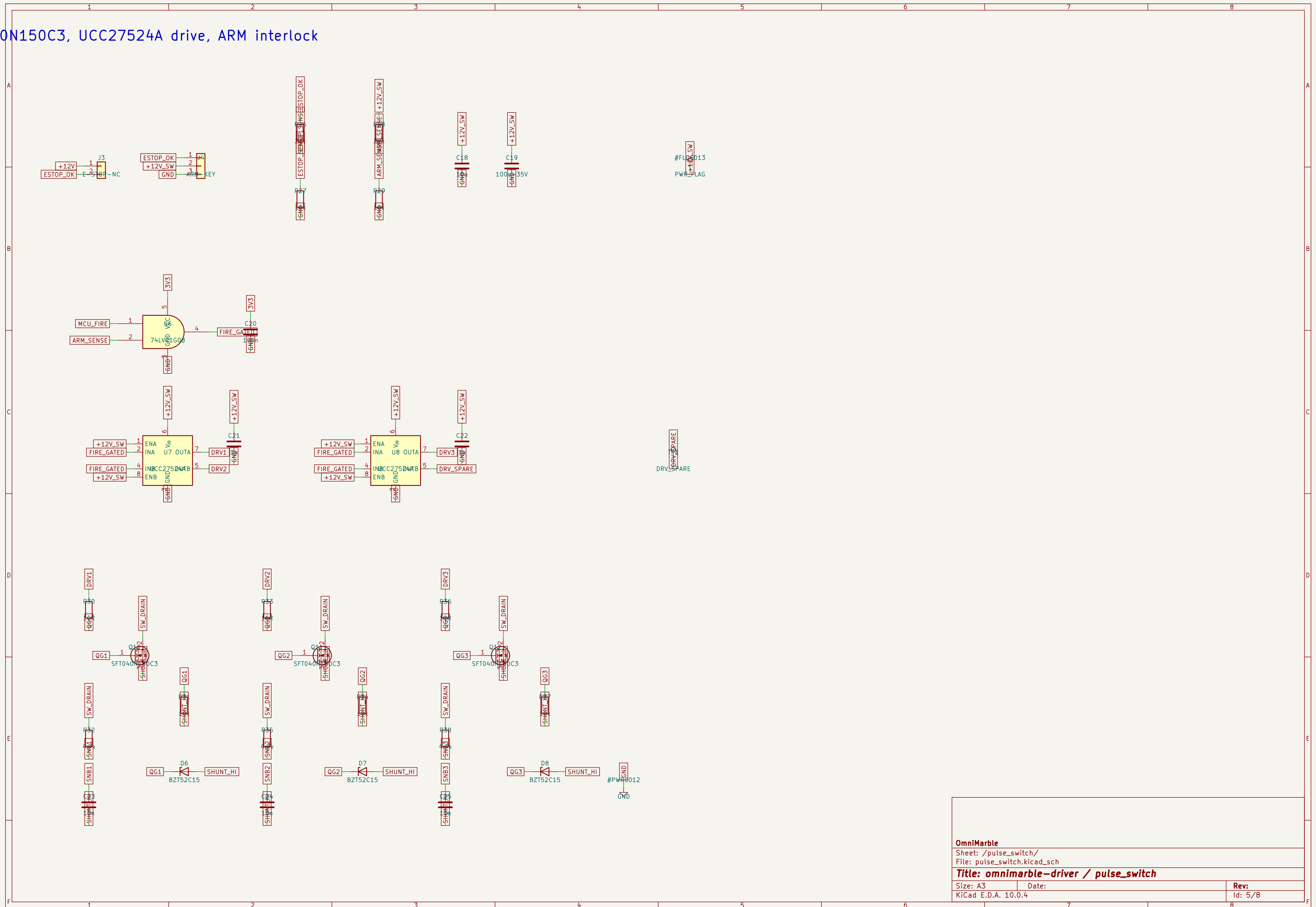
Rev:

KiCad E.D.A. 10.0.4

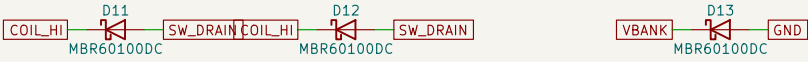
Id: 3/8

2200uF/63V snap-in (populate per bank table)

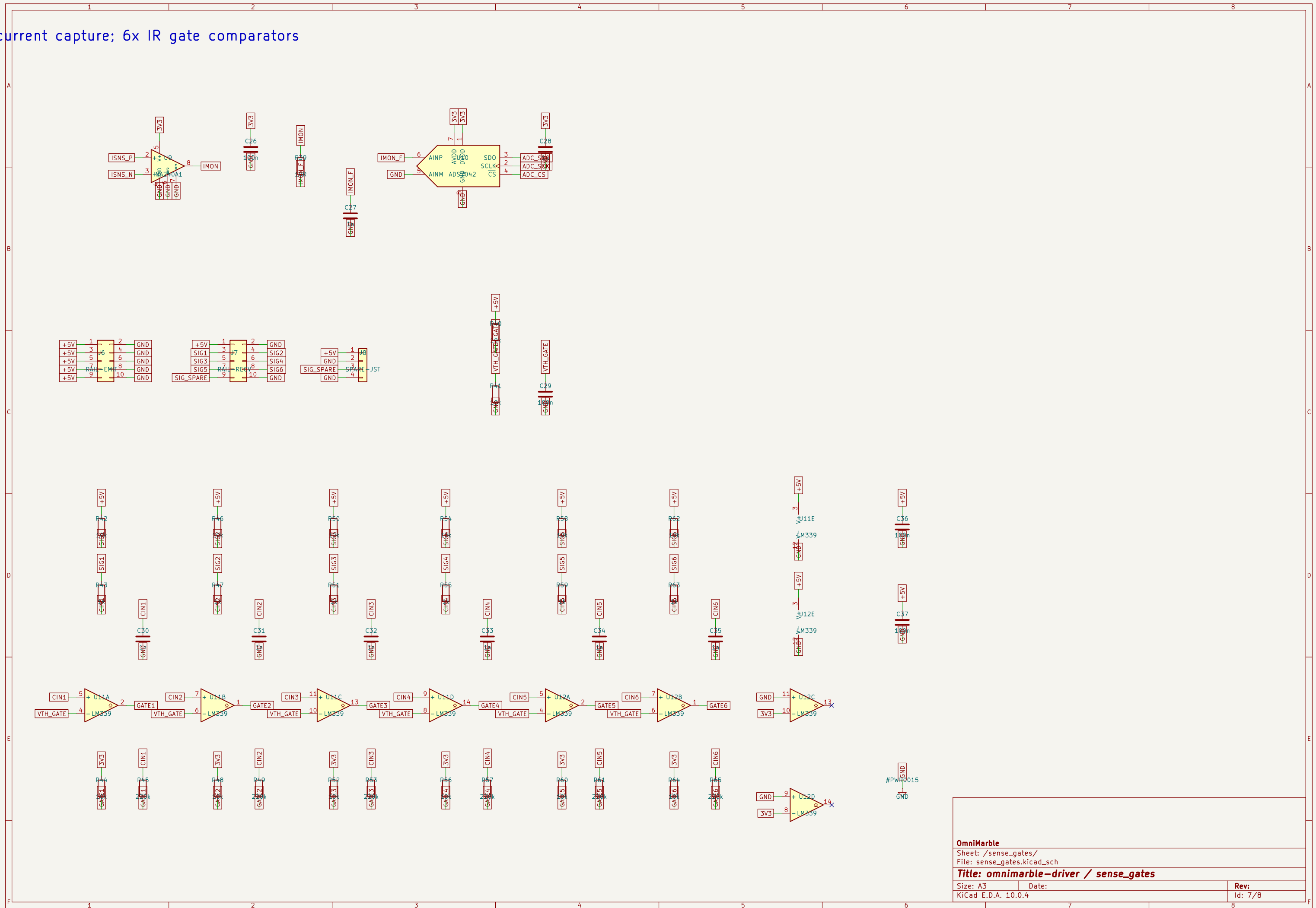
T040N150C3, UCC27524A drive, ARM interlock



blocking -> COIL -> switch -> shunt -> GND



42 current capture; 6x IR gate comparators



0), MCU-agnostic breakout, relay/status drive

